

ESP32-P4-WIFI6-Touch-LCD-4B

Onboard camera interface

Onboard Ethernet port + RS485 + 2-ch relay



This series is a high-performance development board based on the ESP32-P4 chip. This series adopts a dual-core plus single-core RISC-V architecture design, featuring a built-in 4inch 720 × 720 resolution IPS touch display. It supports rich human-machine interaction interfaces, including MIPI-CSI with an integrated image signal processor (ISP), and USB OTG 2.0 HS port for high-speed connections. The ESP32-P4 chip integrates a digital signature peripheral and a dedicated key management unit to ensure data and operational security. It is designed for high performance and high security applications, and fully meets the higher requirements of embedded applications for human-computer interface support, edge computing capabilities and IO connection features.

SKU	Product
31416	ESP32-P4-WIFI6-Touch-LCD-4B



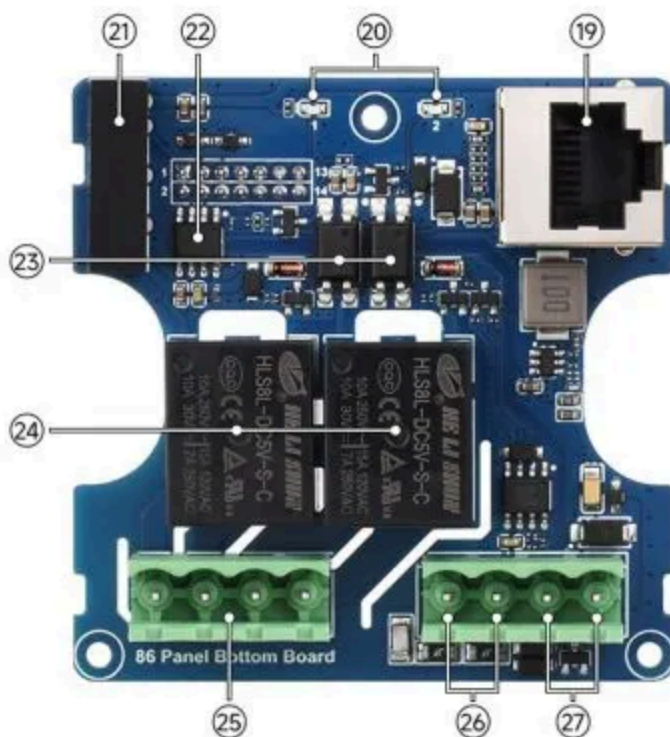
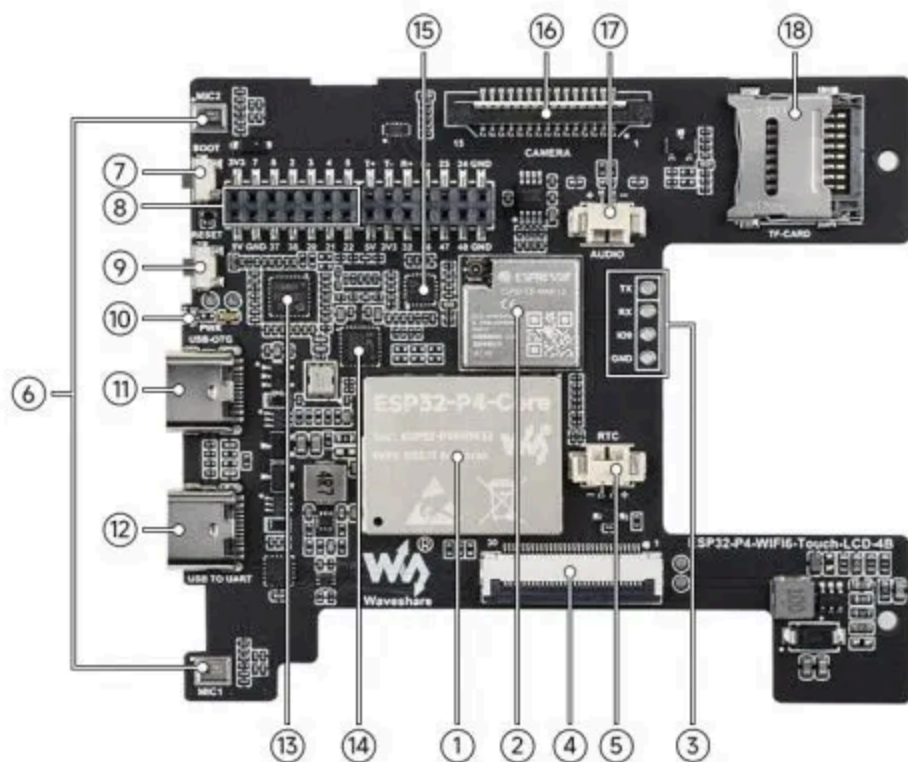
SKU	Product
31570	ESP32-P4-86-Panel-ETH-2RO

Features

- Processor
 - Equipped with a RISC-V 32-bit dual-core processor (HP system), featuring DSP and ISA extensions, a Floating-Point Unit (FPU), with a main frequency of up to 360 MHz.
 - Equipped with a RISC-V 32-bit single-core processor (LP system), with a main frequency of up to 40 MHz.
 - Equipped with an ESP32-C6 WIFI/BT co-processor, expanding WIFI 6/Bluetooth 5 and other functions through SDIO
- Memory
 - 128 KB of high-performance (HP) system read-only memory (ROM).
 - 16 KB of low-power (LP) system read-only memory (ROM).
 - 768 KB of high-performance (HP) L2 memory (L2MEM).
 - 32 KB of low-power (LP) SRAM.
 - 8 KB of system tightly coupled memory (TCM).
 - 32 MB PSRAM stacked in the package, plus 32 MB Nor Flash externally
- Peripheral Interfaces
 - Onboard interfaces include MIPI-CSI, MIPI-DSI, USB 2.0 OTG, Ethernet, SDIO 3.0 TF card slot, dual microphones with echo cancellation, speaker terminal, RTC battery terminal, etc. (different products expose different subsets; refer to the resource introduction)

Hardware Description



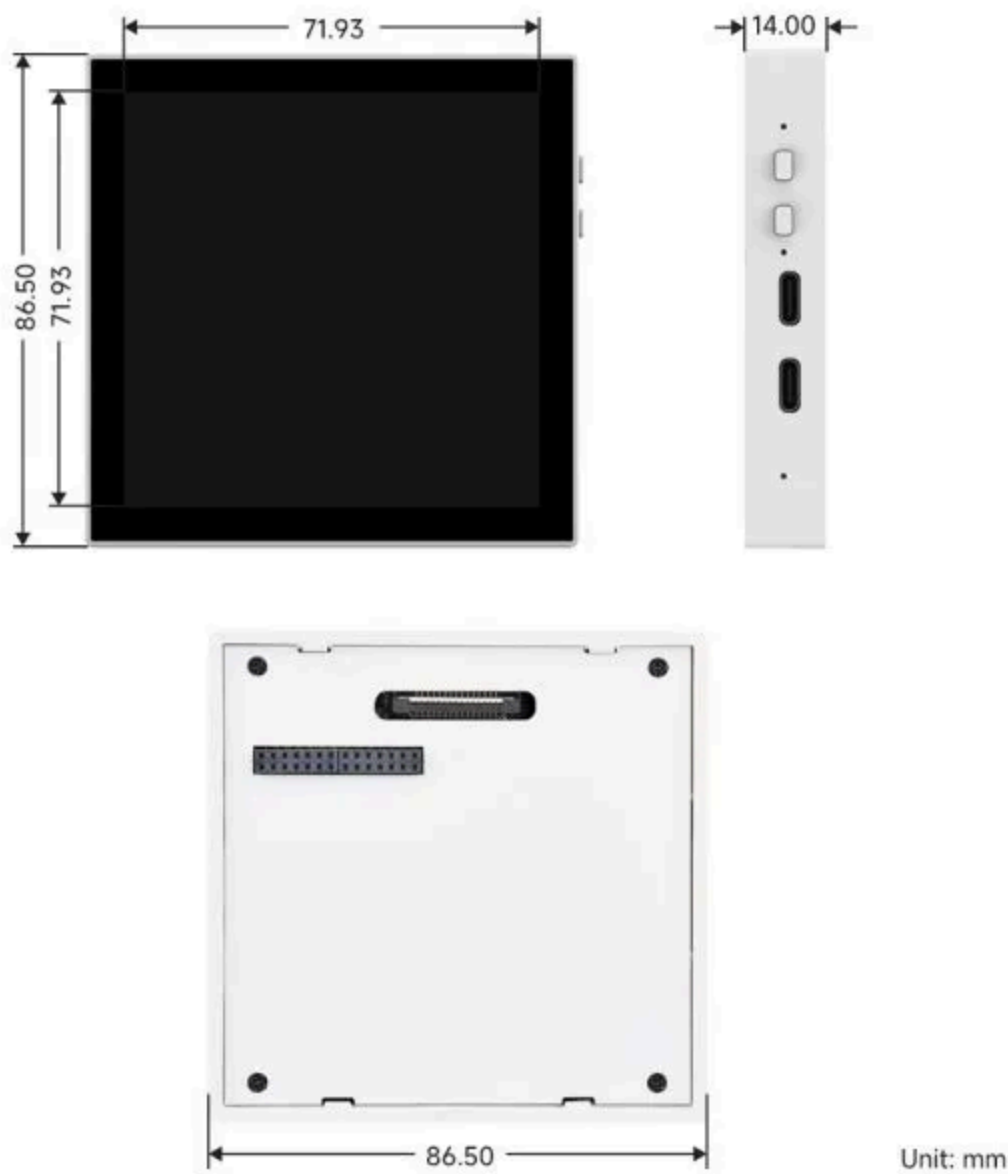


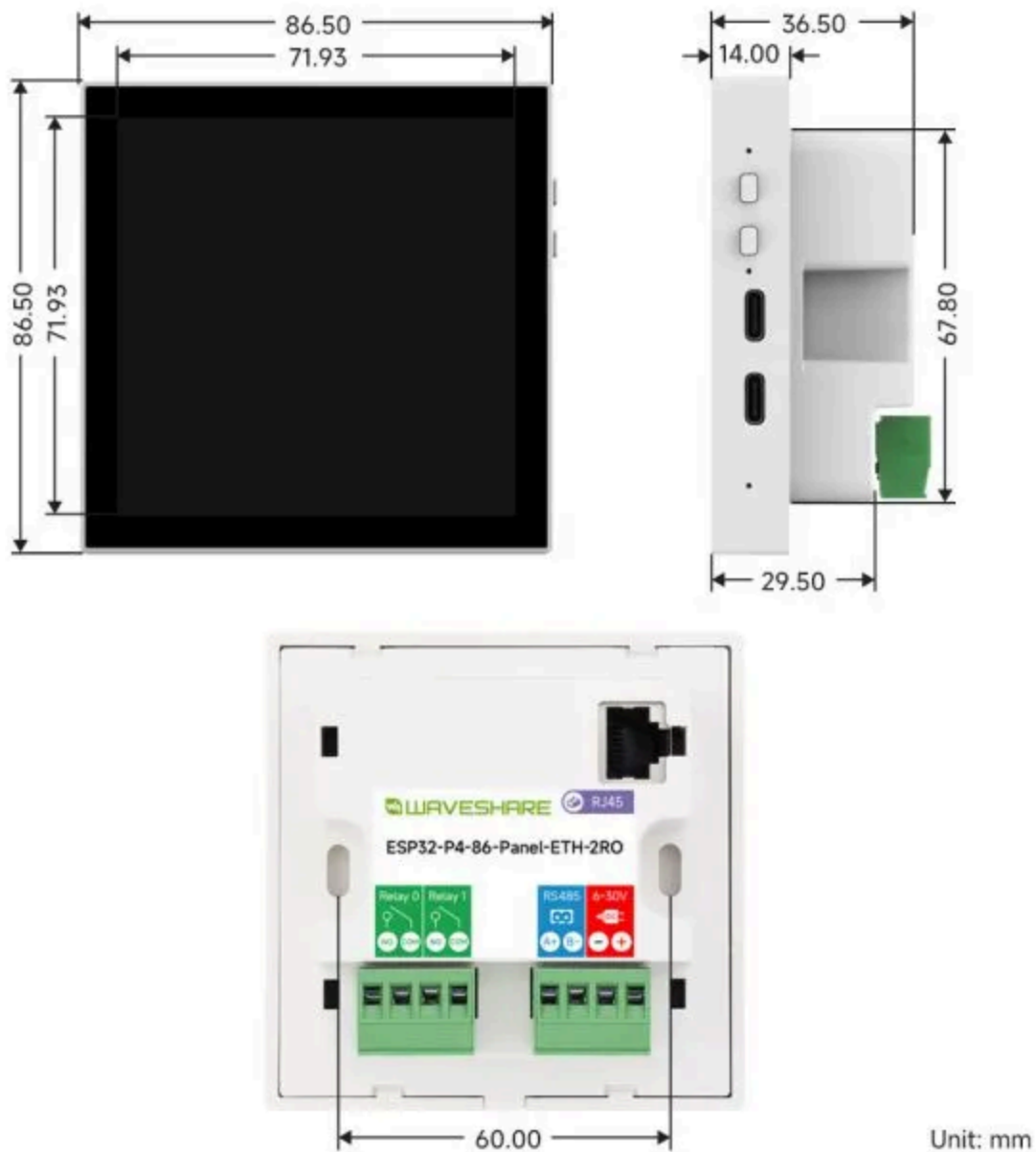
1. **ESP32-P4-Core** Integrates ESP32-P4NRW32 and 32MB Nor Flash

2. **ESP32-C6-MINI-1U-H8 Module** SDIO interface protocol, expands the development board with Wi-Fi 6, Bluetooth 5 (LE)
3. **2.54mm 4PIN Pads** For flashing firmware to the ESP32-C6 module
4. **MIPI DSI LCD IOnterface** For connecting the MIPI 4inch screen
5. **RTC Battery Header** For connecting a rechargeable RTC battery (only supports rechargeable RTC batteries)
6. **Onboard Microphone** Microphone input and echo cancellation
7. **BOOT Button** Press during power-up or reset to enter download mode
8. **Expansion Header (2.0mm pitch)** Can be used to connect the 86 Panel Bottom Board or external I/O
9. **RESET Button**
10. **PWR LED Indicator** Power indicator
11. **Type-C Interface (USB OTG)** USB OTG 2.0 High Speed interface
12. **Type-C Interface (USB TO UART)** For power supply, programming, and debugging
13. **ES7210 Echo Cancellation Chip** Used for echo cancellation to improve audio capture accuracy
14. **IP101** Ethernet port chip
15. **ES8311 Low-Power Audio Codec Chip** For audio encoding and decoding
16. **MIPI CSI Interface** 15PIN, 1.0mm pitch, supports MIPI-2lane cameras (only available on ESP32-P4-WIFI6-Touch-LCD-4B version)
17. **Speaker Header** MX1.25 2P connector, supports 8Ω 2W speaker
18. **TF Card Slot** SDIO 3.0 interface protocol
19. **100M RJ45 Ethernet Port**
20. **Relay Indicators**
21. **Power isolation module**
22. **Digital isolator**
23. **Optocoupler isolation**
24. **Relay**
25. **Relay screw terminal** 5.08mm 4PIN
26. **RS485 screw terminal**
27. **Power screw terminal** DC 6V~30V wide voltage range



Dimensions





Development Tools

Each of these two development approaches has its own advantages, and developers can choose according to their needs and skill levels. Arduino is suitable for beginners and non-professionals because they are easy to learn and quick to get started. ESP-IDF is a better choice for developers with a professional background or high performance requirements, as it provides more advanced development tools and greater control capabilities for the development of complex projects.



WARNING



Due to the limited support for ESP32-P4 on the Arduino platform, ESP-IDF is currently recommended for development.

- **ESP-IDF**, short for Espressif IoT Development Framework, is a development framework provided by Espressif for the ESP series chips. It is developed using the C language, including a compiler, debugger, and flashing tool, etc., and can be developed via the command lines or through an integrated development environment (such as Visual Studio Code with the Espressif IDF plugin). The plugin offers features such as code navigation, project management, and debugging, etc. We recommend using VS Code for development. For the specific configuration process, please refer to the **Working with ESP-IDF**. The tutorial also provides relevant example programs for reference.

 Give Feedback

